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To the Editors,
[Journal name]

Re: "An Exact Decomposition of Paired Score Differences: Separating Prior Fit from Within-Group Resolution" — new submission

Dear Editors,

Please consider the enclosed manuscript for publication in *[Journal name]*.

The problem, in one paragraph. Two probabilistic classifiers are compared on a test set and the newer one scores higher. If the labels depend partly on a grouping variable that both models observe, the score difference sums two distinct improvements: the new model may fit the label frequencies *within* groups more closely, or it may match its predictions to the individual cases they were made for more closely. These are different achievements — the first is skill a lookup table also supplies and that a shift in label frequencies removes; the second is not — and an aggregate score difference cannot tell them apart. Neither can a significance test on that difference, since both components are present in the same number.

The contribution. The paper decomposes the difference exactly, using no fitted reference model. The construction is elementary: because both models emit a full probability vector for every test case, each model's score can be recomputed against any label, so one can score a case's predictions against a *different* case's label from the same group. That relabelling preserves the group and its label frequencies while breaking the correspondence between a prediction and the case it was made for. The gain that survives is prior fit; the gain that disappears is case-specific. The resulting identity is exact in the observed sample, for any score, with no held-out fold and no tuning parameter, at a cost of $O(NK)$ in N cases and K labels.

The remainder is a U-statistic of order two, and it is not a new quantity: it equals the within-group covariance between the two models' predicted-probability difference and the label indicator, which identifies it as the resolution term of the classical reliability–resolution decomposition of a proper score (Murphy 1973; DeGroot & Fienberg 1983; Bröcker 2009), evaluated for a *pair* of forecasters instead of one. What the classical decomposition does not supply is inference for the pairwise contrast, and that is where the paper's technical content sits. Specifically, we prove: the covariance identity for the whole Bregman-score family, so the quadratic and logarithmic scores are named cases of one theorem; asymptotic normality of the dataset-level estimator with a consistent cluster-robust sandwich variance; an exact reduction, at a trivial grouping variable, to a clustered Diebold–Mariano/Giacomini–White test, which locates the estimator precisely relative to comparative forecast evaluation; an exact $(n_c - 1)/n_c$ attenuation of the in-sample plug-in, which is why no sample splitting is needed; an explicit between-group covariance term for what happens when groups are merged; and a Cauchy–Schwarz sensitivity bound, in the tradition of Rosenbaum and Cinelli & Hazlett, for how far an unrecorded grouping variable could move the result.

Validation, and one application in detail. Because the object is a measuring instrument, the paper validates it before using it: signal recovery, null calibration, coverage of the primary interval in the clustered small-group regime that stresses it, and discriminant-validity designs establishing what the two components do *not* separate. One of those is a genuine limitation rather than a strength — a monotone recalibration moves score between the components while adding no information — and the paper responds with a calibration-matching protocol that is part of the procedure, plus a demonstration that it works.

The main application audits two publicly released scene-graph checkpoints of a widely cited debiasing method, evaluated in both modes with the original authors' code, so the models are not ours and the claimed improvement is one the field already reports. Once confidence is matched on held-out images, the proper-score difference is overwhelmingly prior fit under both the quadratic and the logarithmic score, although the two scores disagree on whether any resolution remainder survives (-0.00006, 95% CI $[-0.00120, +0.00109]$ under the quadratic score; $+0.04396$, CI $[+0.04079, +0.04714]$ under the log score) — in either reading at least an order of magnitude smaller than the prior-side shift. A method therefore buys ten points of mean recall while changing within-group resolution by at most a small fraction of that, possibly none. Since its construction subtracts a context-only prediction, this characterises the method rather than indicting it, while showing that a metric read as evidence about rare categories can move sharply on evidence that is mostly, and perhaps entirely, of a different kind.

Three further studies — our own relation predictors, a frozen-CLIP variant of them, and long-tailed classification in images and in text — are reported in an appendix, and include a case in which a model that loses on every aggregate measure is shown to resolve individual cases significantly better than the model it appears to lose to.

Fit. The central contribution is an estimand together with its exact finite-sample and asymptotic inference theory. The benchmarks are where the instrument is exercised and validated, not the claim: no state-of-the-art result is asserted, and none is needed for the argument. I have written the paper so that the contribution is legible before any notation appears — a plain statement of the identity in the introduction, a four-point worked example that exhibits every property proved later, and a table fixing the paper's vocabulary against its standard statistical counterparts.

All code, cached model outputs and scripts needed to reproduce every number, table and figure are publicly available at `https://github.com/vinhqdang/WAGER`, including a verification script that checks each reported value against the committed results files.

The manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. An earlier and substantially different version was submitted to another journal and declined; that submission is closed. I am the sole author and approve this submission. I declare no competing financial or personal interests, and the work received no specific grant from any funding agency. In accordance with journal policy, the manuscript includes a declaration of generative AI use in its preparation.

Thank you for your consideration.

Sincerely,
Quang-Vinh Dang